

## RITSEM, Sweden

WMO: 020130

Lat: 67.7269N

Lon: 17.4711E

Elev: 543

StdP: 94.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -24.3      | -22.3 | -27.4                         | 0.3 | -24.1 | -25.3 | 0.4 | -22.0 | 12.1                  | -4.8 | 10.7 | -5.9 | 1.0                   | 30   | 0.754 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.8                    | 21.1            | 13.8 | 18.9 | 12.9 | 16.9 | 12.0 | 14.8                | 19.2 | 13.7 | 17.7 | 12.7 | 16.0 | 2.2                  | 170  |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 12.7                            | 9.8 | 16.5 | 11.8 | 9.2 | 15.5 | 10.8 | 8.6 | 14.6 | 43.2          | 19.3 | 40.1 | 17.8 | 37.3 | 16.1 | 18.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.1           | 7.8  | 6.6 | DB                         | -27.0 | 23.6               | 1.7 | 1.9                                                | -28.3 | 25.0       | -29.3 | 26.1       | -30.2 | 27.2       | -31.5 | 28.6 | (4) |
| (5)               |      |     | WB                         | -27.3 | 16.3               | 1.8 | 1.1                                                | -28.5 | 17.1       | -29.6 | 17.7       | -30.5 | 18.4       | -31.8 | 19.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 0.2         | -9.7  | -9.7  | -7.2  | -2.3  | 3.3   | 8.4   | 12.1  | 10.9  | 6.4   | 0.5   | -4.1  | -6.6 | (6)  |  |
|                                                                        | DBStd       | 8.87        | 6.37  | 6.31  | 5.40  | 4.09  | 3.82  | 3.60  | 3.32  | 3.20  | 3.20  | 3.83  | 4.55  | 5.75 | (7)  |  |
|                                                                        | HDD10.0     | 3736        | 611   | 551   | 532   | 368   | 213   | 74    | 16    | 27    | 113   | 293   | 424   | 514  | (8)  |  |
|                                                                        | HDD18.3     | 6609        | 869   | 784   | 791   | 618   | 467   | 299   | 196   | 231   | 357   | 551   | 674   | 772  | (9)  |  |
|                                                                        | CDD10.0     | 169         | 0     | 0     | 0     | 0     | 3     | 25    | 80    | 55    | 6     | 0     | 0     | 0    | (10) |  |
|                                                                        | CDD18.3     | 2           | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0     | 0     | 0     | 0    | (11) |  |
|                                                                        | CDH23.3     | 12          | 0     | 0     | 0     | 0     | 0     | 2     | 8     | 2     | 0     | 0     | 0     | 0    | (12) |  |
|                                                                        | CDH26.7     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.8         | 2.7   | 2.6   | 2.7   | 2.8   | 2.9   | 2.7   | 2.5   | 2.5   | 2.8   | 3.0   | 2.9   | 2.9  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 517         | 52    | 39    | 35    | 22    | 30    | 38    | 62    | 55    | 47    | 49    | 38    | 49   | (15) |  |
|                                                                        | PrecMax     | 626         | 173   | 90    | 80    | 46    | 62    | 56    | 140   | 104   | 102   | 104   | 93    | 126  | (16) |  |
|                                                                        | PrecMin     | 425         | 2     | 14    | 9     | 6     | 9     | 5     | 23    | 23    | 13    | 12    | 7     | 10   | (17) |  |
|                                                                        | PrecStd     | 58          | 34    | 20    | 20    | 12    | 12    | 14    | 32    | 21    | 25    | 20    | 25    | 27   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 3.1   | 3.3   | 4.2   | 8.5   | 18.7  | 21.4  | 24.6  | 22.6  | 16.1  | 10.9  | 6.1   | 3.9  | (19) |  |
|                                                                        |             | MCWB        | 1.8   | 2.0   | 1.4   | 4.4   | 10.3  | 12.4  | 15.5  | 14.3  | 11.0  | 8.1   | 4.4   | 2.3  | (20) |  |
|                                                                        | 2%          | DB          | 1.8   | 1.8   | 2.6   | 6.2   | 13.7  | 18.3  | 21.9  | 20.3  | 13.9  | 8.2   | 4.6   | 2.8  | (21) |  |
|                                                                        |             | MCWB        | 0.6   | 0.3   | 0.9   | 3.2   | 8.2   | 11.1  | 14.6  | 13.7  | 10.1  | 6.3   | 3.3   | 1.4  | (22) |  |
|                                                                        | 5%          | DB          | 0.6   | 0.6   | 1.3   | 4.6   | 10.9  | 16.0  | 19.6  | 17.9  | 12.3  | 7.0   | 3.0   | 1.7  | (23) |  |
|                                                                        |             | MCWB        | -0.4  | -0.5  | -0.2  | 2.2   | 6.5   | 10.3  | 13.6  | 13.0  | 9.5   | 5.6   | 1.7   | 0.4  | (24) |  |
|                                                                        | 10%         | DB          | -1.0  | -0.8  | 0.0   | 3.2   | 8.5   | 14.3  | 17.5  | 16.0  | 11.1  | 5.7   | 1.7   | 0.5  | (25) |  |
|                                                                        |             | MCWB        | -1.9  | -1.8  | -1.4  | 1.2   | 5.0   | 9.5   | 12.6  | 12.0  | 8.7   | 4.2   | 0.5   | -0.8 | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 2.1   | 2.2   | 2.1   | 4.8   | 10.8  | 13.7  | 16.6  | 15.8  | 12.1  | 8.4   | 4.7   | 2.7  | (27) |  |
|                                                                        |             | MCDB        | 2.8   | 3.2   | 3.4   | 7.5   | 17.6  | 18.2  | 22.7  | 20.8  | 14.8  | 10.6  | 5.9   | 3.6  | (28) |  |
|                                                                        | 2%          | WB          | 0.8   | 0.6   | 1.0   | 3.5   | 8.6   | 12.2  | 15.4  | 14.5  | 10.8  | 6.8   | 3.4   | 1.4  | (29) |  |
|                                                                        |             | MCDB        | 1.6   | 1.5   | 2.3   | 5.9   | 13.3  | 16.4  | 20.4  | 18.4  | 13.1  | 8.2   | 4.5   | 2.4  | (30) |  |
|                                                                        | 5%          | WB          | -0.2  | -0.4  | 0.0   | 2.4   | 6.8   | 11.0  | 14.2  | 13.5  | 9.8   | 5.6   | 1.9   | 0.3  | (31) |  |
|                                                                        |             | MCDB        | 0.6   | 0.5   | 1.3   | 4.4   | 10.6  | 15.5  | 18.6  | 17.2  | 12.0  | 6.9   | 2.9   | 1.3  | (32) |  |
|                                                                        | 10%         | WB          | -2.0  | -1.8  | -1.3  | 1.3   | 5.3   | 9.9   | 13.2  | 12.5  | 8.9   | 4.3   | 0.7   | -1.0 | (33) |  |
|                                                                        |             | MCDB        | -0.9  | -0.8  | -0.1  | 3.0   | 8.2   | 13.9  | 16.7  | 15.7  | 10.9  | 5.5   | 1.6   | 0.1  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.2   | 6.1   | 6.2   | 5.9   | 5.7   | 6.4   | 6.8   | 6.3   | 5.2   | 4.2   | 4.7   | 5.4  | (35) |  |
|                                                                        |             | MCDBR       | 6.0   | 6.0   | 5.7   | 6.0   | 9.5   | 9.8   | 10.2  | 9.3   | 7.0   | 4.8   | 4.9   | 5.5  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.4   | 5.5   | 4.7   | 4.2   | 5.3   | 5.2   | 5.0   | 4.6   | 4.4   | 3.8   | 4.0   | 4.7  | (37) |  |
|                                                                        |             | MCWBR       | 5.6   | 6.2   | 5.7   | 5.7   | 9.0   | 8.9   | 9.1   | 8.4   | 6.2   | 4.8   | 4.6   | 5.5  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.167 | 0.227 | 0.268 | 0.292 | 0.305 | 0.311 | 0.320 | 0.308 | 0.281 | 0.242 | 0.171 | N/A  | (40) |  |
|                                                                        |             | taud        | 1.924 | 2.141 | 2.213 | 2.249 | 2.316 | 2.464 | 2.517 | 2.531 | 2.562 | 2.279 | 1.958 | N/A  | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 159   | 607   | 781   | 863   | 890   | 895   | 875   | 849   | 793   | 585   | 135   | 0    | (42) |  |
|                                                                        |             | Edh at Noon | 9     | 45    | 77    | 99    | 104   | 93    | 86    | 75    | 56    | 39    | 8     | 0    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.04  | 0.39  | 1.63  | 3.58  | 4.87  | 4.90  | 4.43  | 3.28  | 1.72  | 0.57  | 0.10  | 0.00 | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.00  | 0.08  | 0.25  | 0.26  | 0.43  | 0.57  | 0.48  | 0.36  | 0.18  | 0.11  | 0.01  | 0.00 | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page